

Assignment Alpha

BAI-7A

23K-0055

Q1) 1. Total pixels = $1280 \times 720 = 921600$

$$\text{Bits per frame} = \text{pixels} \times X$$

$$29491200 = 921600 X$$

$$X = \frac{29491200}{921600}$$

$$X = 32$$

2. ~~29491200~~ $\frac{29491200}{8} = 3686400 \text{ bytes/frame}$

3. $29491200 \times 30 = 884736000 \text{ bits/s}$

Q2) $Q = \begin{bmatrix} 1.5 & 3 \\ 2 & 4 \\ 2.2 & 4.4 \\ 2.4 & 4.4 \end{bmatrix}$

$$K = \begin{bmatrix} 1.1 & 2.2 \\ 1 & 3 \\ 1 & 3 \\ 1.1 & 3 \end{bmatrix}$$

Query 1: AI

Query 2: Governance

w/ AI key

$$[1.5, 3] \cdot [1.1, 2.2]$$

$$= 1.65 + 6.6$$

$$= 8.25$$

w/ AI key

$$[2, 4] \cdot [1.1, 2.2]$$

$$= 2.2 + 8.8$$

$$= 11$$

w/ Governance key

$$[1.5, 3] \cdot [1, 3]$$

$$= 1.5 + 9$$

$$= 10.5$$

w/ Governance key

$$[2, 4] \cdot [1, 3]$$

$$= 2 + 12$$

$$= 14$$

w/ Emerging key

$$[1.5, 3] \cdot [1, 3]$$

$$= 1.5 + 9$$

$$= 10.5$$

w/ Emerging key

$$[2, 4] \cdot [1, 3]$$

$$= 2 + 12$$

$$= 14$$

w/ Course key

$$[1.5, 3] \cdot [1.1, 3]$$

$$= 1.65 + 9$$

$$= 10.65$$

w/ Course key

$$[2, 4] \cdot [1.1, 3]$$

$$= 2.2 + 12$$

$$= 14.2$$

$$[8.25, 10.5, 10.5, 10.65]$$

$$[11, 14, 14, 14.2]$$

Query 3: Emerging

w/ AI key

$$\begin{aligned} & [2.2, 4.4] \cdot [1.1, 2.2] \\ &= 2.42 + 9.68 \\ &= 12.1 \end{aligned}$$

w/ Governance key & Emerging key

$$\begin{aligned} & [2.2, 4.4] \cdot [1, 3] \\ &= 2.2 + 13.2 \\ &= 15.4 \end{aligned}$$

Course
w/ ~~Emerging~~ key

$$\begin{aligned} & [2.2, 4.4], [1.1, 3] \\ &= 2.42 + 13.2 \\ &= 15.62 \end{aligned}$$

[12.1, 15.4, 15.4, 15.62]

Query 4: Course

w/ AI key

$$\begin{aligned} & [2.4, 4.4] \cdot [1.1, 2.2] \\ &= 2.64 + 9.68 \\ &= 12.32 \end{aligned}$$

w/ Governance key & Emergence key

$$\begin{aligned} & [2.4, 4.4] \cdot [1, 3] \\ &= 2.4 + 13.2 \\ &= 15.6 \end{aligned}$$

w/ Course key

$$\begin{aligned} & [2.4, 4.4] \cdot [1.1, 3] \\ &= 2.64 + 13.2 \\ &= 15.84 \end{aligned}$$

[12.32, 15.6, 15.6, 15.84]

$$QK^T = \begin{bmatrix} 8.25 & 10.5 & 10.5 & 10.65 \\ 11 & 14 & 14 & 14.2 \\ 12.1 & 15.4 & 15.4 & 15.62 \\ 12.32 & 15.6 & 15.6 & 15.84 \end{bmatrix}$$

$$d_k = 2$$

$$\frac{QK^T}{\sqrt{d_k}} = \begin{bmatrix} 5.834 & 7.425 & 7.425 & 7.531 \\ 7.778 & 9.899 & 9.899 & 10.041 \\ 8.556 & 10.889 & 10.889 & 11.045 \\ 8.712 & 11.031 & 11.031 & 11.201 \end{bmatrix}$$

Applying softmax row-wise, we have the attention-weight matrix

$$A = \begin{bmatrix} 0.0614 & 0.3016 & 0.3016 & 0.3354 \\ 0.0366 & 0.3056 & 0.3056 & 0.3521 \\ 0.0297 & 0.3063 & 0.3063 & 0.3578 \\ 0.0300 & 0.3046 & 0.3049 & 0.3609 \end{bmatrix}$$

Multiplying A w/ V, we have

$$\text{Attention}(Q, K, V) = \begin{bmatrix} 2.0168 & 0.3000 \\ 2.0656 & 0.3000 \\ 2.0794 & 0.3000 \\ 2.0792 & 0.3000 \end{bmatrix}$$

$$Q3) 1) P(\text{Yes}) = \frac{3}{5} = 0.6$$

$$P(\text{No}) = \frac{2}{5} = 0.4$$

Conditional Probabilities

{ Weak, Strong }

For Rain = Yes

$$P(\text{Weak} | \text{Yes}) = \frac{2+1}{3+2} \Rightarrow \frac{3}{5} = 0.6$$

~~Strong~~

$$P(\text{Strong} | \text{Yes}) = \frac{1+1}{5} = 0.4$$

$$P(\text{High} | \text{Yes}) = \frac{2+1}{5} = 0.6$$

$$P(\text{Low} | \text{Yes}) = \frac{1+1}{5} = 0.4$$

For Rain = No

$$P(\text{Weak} | \text{No}) = \frac{0+1}{2+2} \Rightarrow 0.25$$

$$P(\text{Low} | \text{No}) = \frac{2+1}{4} = 0.75$$

Rain Yes

$$0.6 \times 0.6 \times 0.6 = 0.216$$

$$0.216 > 0.05$$

Rain No

$$0.4 \times 0.5 \times 0.25 = 0.05$$

so Rain = Yes

$$\text{Normalized: } P(\text{Yes} | \text{Weak, High}) = \frac{0.216}{0.216 + 0.05} \Rightarrow 0.812$$

Strong + High

$$\text{Yes: } 0.6(0.4)(0.6) \Rightarrow 0.144$$

$$\text{No: } 0.4(0.5)(0.25) = 0.05$$

Weak + Low

$$\text{Yes: } 0.6(0.6)(0.4) = 0.144$$

$$\text{No: } 0.4(0.5)(0.75) = 0.15$$

Rain = Yes

Rain = No

Strong + Low

$$\text{Yes: } 0.6(0.4)(0.4) = 0.096$$

$$\text{No: } 0.4(0.5)(0.75) = 0.15$$

Rain = No

2) Entropy

$$H(\text{rain}) \Rightarrow -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.971$$

IG of wind

Wind = Weak

$$H(\text{Weak}) = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.918$$

Wind = Strong

$$H(\text{Strong}) = 1$$

$$H(\text{rain} | \text{Wind}) = \frac{3}{5}(0.918) + \frac{2}{5}(1) = 0.951$$

$$IG(\text{Wind}) = 0.971 - 0.951 = 0.020$$

IG of Humidity

$$H(\text{High}) = 0$$

Humidity = Low

$$H(\text{Low}) = 0.918$$

$$H(\text{Rain} | \text{Humidity}) = \frac{2}{5}(0) + \frac{3}{5}(0.918) = 0.551$$

$$IG(\text{Humidity}) = 0.971 - 0.551 \approx 0.420$$

$$IG(\text{Humidity}) > IG(\text{Wind})$$

- Humidity should become the root node of a decision tree

Logistic Regression

$$b = w_1 = w_2 = 0$$

$$\sigma(0) = \frac{1}{1+e^0} = 0.5$$

$$\hat{y} = \{0.5, 0.5, 0.5, 0.5, 0.5\}$$

$$L = -\ln(0.5)$$

$$= 0.6931$$

bias gradient

$$\frac{\partial L}{\partial b} = \frac{1}{5} \sum (\hat{y} - y)$$

$$= \frac{(0.5-1) + (0.5-1) + (0.5-0) + (0.5-1) + (0.5-0)}{5}$$

$$\frac{\partial L}{\partial b} = -0.1$$

For wind

$$\frac{\partial L}{\partial w_1} = \frac{(-0.5) + 0.5}{5} = 0$$

For humidity

$$\frac{\partial L}{\partial w_2} = \frac{-0.5 - 0.5}{5} = -0.2$$

$$\alpha = 0.1$$

$$w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial L}{\partial w}$$

$$b = 0 - (0.1)(-0.1) = 0.01$$

$$w_1 = 0 - (0.1)(0) = 0$$

$$w_2 = 0 - (0.1)(-0.2) = 0.02$$

4) All three methods extract different information. Naive Bayes uses conditional probabilities & provides direct Rain Yes/No predictions. Information Gain shows that Humidity is considerably more informative than wind, making Humidity the first decision-tree feature. Logistic regression instead learns numerical weights & estimates probability of rain through sigmoid function

(Q5) 1. Joint probability of both being infected

infection is independent & both fields have equal probability

$$P(A=I) = 0.5$$

$$P(B=I) = 0.5$$

$$P(A \cap B) = P(A)P(B) = 0.25$$

2.	Time	Location	Percept	Action	Credit	Total Credit
	0	A	Infected	Spray A	\$0	\$0
	1	B	Infected	Spray B	\$0	\$0
	2	A	Healthy	Move to B	\$2	\$2
	3	B	Healthy	Move to A	\$2	\$4

Total Credit = \$4

(Q7) 1. $A = (6, -2, \sqrt{9})$

$$A = (6, -2, 3)$$

$$b = (4, X, -2)$$

$A \cdot b = 0$ if both vectors are perpendicular

$$(6)(4) + (-1)(X) + (3)(-2) = 0$$

$$24 - X - 6 = 0$$

$$18 - X = 0$$

$$\boxed{X = 18}$$

2. Dot product multiplies corresponding elements and add them

The answer is a single scalar.

Hadamard product multiplies elements but do not add them
so the answer becomes another vector

3. $u = (1, 2)$

$$v = (2, 2)$$

$$u \cdot v = (1)(2) + 2(2)$$

$$u \cdot v \Rightarrow 2 + 4 = 6$$

using L2 norm:

$$\|u+v\|_2^2 = \|u\|_2^2 + \|v\|_2^2 + 2(u \cdot v)$$

$$\|u+v\|_2 = \sqrt{3^2 + 4^2}$$

$$\|u+v\|_2 = \sqrt{25}$$

$$\|u+v\|_2^2 = (\sqrt{25})^2 = 25$$

$$\|u\|_2 = \sqrt{5}$$

$$\|v\|_2 = \sqrt{8}$$

$$\|u\|_2^2 = 5$$

$$\|v\|_2^2 = 8$$

~~∴ $\|u+v\|_2^2$~~

$$\therefore 25 = 5 + 8 + 2(u \cdot v)$$

$$25 = 13 + 2(u \cdot v)$$

$$12 = 2(u \cdot v)$$

$$\boxed{u \cdot v = 6} \text{ proven.}$$

using vector projection

$$\text{Proj}_v(u) = \frac{u \cdot v}{\|v\|^2} v$$

$$\rightarrow \text{Proj}_v(u) = \frac{6}{8} (2, 2)$$

$$= \frac{3}{4} (2, 2)$$

$$u \cdot v = 6$$

$$= (1.5, 1.5)$$

$$\|v\|^2 = 8$$

$$(1.5, 1.5) = \frac{u \cdot v}{8} (2, 2)$$

~~Proven~~

$$1.5 = \frac{u \cdot v}{8} (2)$$

$$12 = 2(u \cdot v)$$

$$\boxed{u \cdot v = 6} \text{ proved.}$$

(Q 8) $S(t) = 0.2t + t^2 + \sin(t) + \cos(t) + t^5 + t^6 + \sin(t) + t^5 + t^8 + \sin(t) + \cos(t)$

$$S(t) = 0.2t + t^2 + 2t^5 + t^6 + t^8 + 3\sin(t) + 2\cos(t)$$

At $t=1$, $S(1) = 0.2 + 1 + 2 + 1 + 1 + 3\sin(1) + 2\cos(1)$

$$S(1) = 7.252$$

At $t=2$, $S(2) = 0.2(2) + 2^2 + 2(2^5) + 2^6 + 2^8 + 3\sin(2) + 2\cos(2)$

$$S(2) \Rightarrow 390.503$$

At $t=3$, $S(3) = 0.2(3) + 3^2 + 2(3^5) + 3^6 + 3^8 + 3\sin(3) + 2\cos(3)$

$$S(3) \Rightarrow 7187.754$$

At $t=4$, $S(4) = 0.2(4) + 4^2 + 2(4^5) + 4^6 + 4^8 + 3\sin(4) + 2\cos(4)$

$$S(4) = 71699.004$$

FAIRGAME paper

Q6) 1. Game theory models situations where each agent's best choice depends on what the other agents do. It gives a theoretical benchmark, the Nash equilibrium, against which LLM agents' actual behavior can be compared. In FAIRGAME, changing only the language or personality while keeping payoffs fixed reveals hidden biases. It also helps design incentives that make cooperation between agents stable.

2. Game theory is a mathematical tool used for strategic decision-making among multiple parties. It explains why rational individuals can end up worse off together, and how coordination and cooperation emerge.

3. The paper uses Prisoner's Dilemma and the Battle of the Sexes. The paper itself does not cover SG, but both games map onto it. Prisoner's Dilemma models power control & spectrum sharing, where aggressive transmission harms everyone. The Battle of the Sexes models channel selection & network-slice agreements between operators.

4. Llama 3.1 405B Instruct

Mistral Large

GPT-4o

Claude 3.5 Sonnet

A semi-independent policies training method w/ shared representation paper

~~Q6) 1.~~

Q6) 1. POMDP is the tuple $(S, A, T, R, \Omega, O, \gamma)$, states S , actions A , transition probabilities $T(s'/s, a)$, reward $R(s, a)$, observations Ω , observation probabilities $O(o/s, a)$ and a discount factor γ . The agent cannot see the true state, so it keeps a belief $b(s)$ over states & updates it after each observation. It maximises the expected discount return $E[\sum \gamma^t r_t]$. The paper uses the multi-agent version, a Dec-POMDP, in which n agents choose a joint action & share one team reward, but each acts only on its own local observation.

2. Hard sharing suits closely related tasks. For example, autonomous driving, multi-task NLP built on BERT, speech recognition combined w/ speaker identification, and this paper's multi-agent RL.

Soft sharing gives each task its own model, with the models regularized to stay similar. It suits loosely related tasks, such as multilingual models, domain adaptation and federated learning. Hard sharing is efficient but rigid; soft sharing is flexible but uses more parameters.

3. It is the loss for the forward model. For each agent i , the decoders f_o and f_r take the agent's type embedding z , its observation o_i and the team reward r . The loss is the squared observation error plus λ_e times the squared reward error, summed over agents and averaged over replay-buffer samples $\mathcal{D}$. Because only z carries agent-specific information into the decoder, z learns to encode what makes each agent type behave differently.

4. Outer loop is the standard RL where agents act, environment returns new states, and agents receive the game reward R_t . In the inner loop, the representation model generates two extra rewards. R^e rewards high prediction error to encourage exploration. R^d rewards same-type agents for having similar representations and different types for having distinct ones, which preserves specialization. All these are summed into the total reward.